

Programme: B E – Information Science & Engineering

Internal Assessment – II

TERM : **15/04/2024 to 22/07/2024**

COURSE NAME : **Database Management Systems**

DATE : **24 /07/2024** TIME: **2PM -3PM**

COURSE CODE : **IS44**

SEMESTER: **IV**

MAX MARKS: **30**

PORTIONS : **L17 – L40**



Mobile Phones are banned

Instructions to Candidates: **Answer any TWO full questions.**

Marks: **15x2=30**

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	List all the six clauses of an SQL retrieval query. Classify them based on mandatory and optional.	L2	CO3	3
b	Discuss insertion, deletion and modification anomalies. Why are they considered bad? Illustrate with examples.	L2	CO4	6
c	What is the Two-phase locking protocol? How does it guarantee serializability?	L2	CO5	6
2.a	List the various aggregation functions that can be used in SQL.	L2	CO3	3
b	Consider the universal relation $R = \{A, B, C, D, E, F, G, H, I\}$ and set of FD's; $\{AB \rightarrow C, A \rightarrow DE, B \rightarrow F, F \rightarrow GH, D \rightarrow IJ\}$ i) What are the candidate keys of R? ii) Decompose R into 2NF and 3NF relations.	L3	CO4	6
c	Explain why concurrency control is needed? Explain with an example any two concurrency problems.	L2	CO5	6
3.a	What are the various purposes and syntax for the ALTER TABLE command? Give an example for each.	L2	CO3	3
	What is Functional Dependency? Write an algorithm to find the minimal cover for set of FD. Construct minimal cover for the following set of FD's $\{B \rightarrow A, D \rightarrow A, AB \rightarrow D\}$.	L3	CO4	6
c	Explain with an example, the ACID Properties of a transaction.	L2	CO5	6

* L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create